

WeatherGPT - SIH Team Work Division

Remote Development Plan for a 6-Member Team

Editable planning document - names can be filled by the team

Recommended structure: only two members work heavily on the prototype code. The other four strengthen testing, presentation, research, validation, and SIH strategy. This reduces integration conflicts when everyone is working from home.

1. Team Structure

Member	Role	Main Responsibility	Primary Deliverable
Person1-shomesh	Prototype - Frontend	WeatherGPT UI, dashboard, maps, charts, responsive screens, data display	Frontend modules and connected UI
Person2-Kannan	Prototype - Backend/AI	n8n, APIs, Gemini, decision logic, backend endpoints, integration support	Working backend workflows and APIs
Person 3-visvaa	Testing & Integration QA	Feature testing, bug tracking, backend/frontend validation, demo readiness	Test checklist, bug log, verified build
Person4-raghul	PPT & Documentation	SIH PPT, architecture diagrams, screenshots, report, demo flow, speaker notes	Presentation package and documentation
Person 5-charudharsini	Research	Problem research, existing solutions, weather/disaster sources, evidence, competitor study	Research notes, references, comparison data
Person 6-ritika	Product Research & SIH Strategy	User needs, uniqueness, feasibility, impact, future scope, judge Q&A	SIH strategy, use cases, Q&A preparation

2. Person 1 - Prototype Frontend

- Build and maintain the WeatherGPT main interface.
- Create dashboard sections for current weather, forecasts, AQI, risk intelligence and activity intelligence.
- Build the Disaster, Map, Copilot, Planner, INTEL+, LAB and System screens.
- Integrate charts, cards, maps, alerts, loading states and error messages.
- Connect frontend components to agreed backend webhook responses.
- Maintain responsive behaviour for laptop and mobile demonstration.
- Avoid changing backend workflow logic unless coordinated with Person 2.

3. Person 2 - Prototype Backend / AI

- Build and maintain the n8n backend workflows.
- Integrate Open-Meteo, NDMA SACHET, NASA EONET and Gemini services.
- Implement weather retrieval, risk logic, activity scoring and decision-support endpoints.

- Maintain stable webhook names and JSON response formats for the frontend.
- Handle Gemini grounding, multilingual support and AI response integration.
- Work closely with Person 1 whenever an API response structure changes.
- Export and preserve working workflow JSON files after major changes.

4. Person 3 - Testing & Integration QA

- Test each feature as soon as a new build is shared; do not wait until the final day.
- Check city search, current weather, hourly and 7-day forecast, AQI, risk, activities, maps, alerts, Gemini, route, campus, event, trip, scenario, heat, languages and voice.
- Test failure cases such as wrong location, backend offline, API failure, Gemini unavailable and no nearby disaster events.
- Maintain a simple bug log with Feature, Problem, Expected Result and Status.
- Re-test every fixed bug and mark it verified.
- Run the complete demo flow before presentation day and identify unstable features.

5. Person 4 - PPT & Documentation

- Maintain the SIH standard presentation continuously instead of creating it on the final day.
- Prepare problem, proposed solution, uniqueness, architecture, workflow, technology stack, impact, feasibility, prototype evidence and future-scope slides.
- Collect clean screenshots from the prototype team.
- Prepare architecture diagrams and simple explanation visuals.
- Maintain the project report and capability summary.
- Prepare a 2-minute, 5-minute and full presentation flow.
- Coordinate with Persons 5 and 6 so research evidence and SIH positioning are reflected in the PPT.

6. Person 5 - Research

- Research the real problem WeatherGPT is solving and collect reliable evidence.
- Study existing systems such as common weather applications, IMD/Mausam, SACHET, Windy and other relevant products.
- Prepare a competitor comparison showing what WeatherGPT adds beyond a normal forecast application.
- Research weather, disaster, accessibility and decision-support use cases.
- Collect credible sources and statistics that can strengthen the SIH presentation.
- Validate technical claims so the team does not exaggerate official warning, forecast or AI capabilities.
- Share short research summaries with the PPT person instead of long raw links only.

7. Person 6 - Product Research & SIH Strategy

- Define the main target users and the strongest practical use cases.
- Prepare examples for students, travellers, event organisers, sports users, disaster-affected users and other relevant groups.
- Identify the project's strongest unique points and explain why it is more than a normal weather application.
- Prepare feasibility, scalability, implementation roadmap and future-scope points.

- Prepare likely judge questions and strong team answers.
- Work with the PPT member on the story: Problem -> Gap -> WeatherGPT -> Demo -> Impact.
- Check whether every major feature contributes to a clear user problem instead of existing only to increase feature count.

8. Prototype Team Collaboration Rule

Person 1 and Person 2 form the core prototype team. They should coordinate frequently because frontend and backend changes directly affect each other. One of these two should maintain the master working build so multiple conflicting copies of WeatherGPT are not created.

9. Minimum Testing Checklist

Test Item	Status	Remarks
Location search	Not Tested	
Current weather	Not Tested	
Hourly forecast	Not Tested	
7-day forecast	Not Tested	
AQI	Not Tested	
Weather risk score	Not Tested	
Activity Intelligence	Not Tested	
Best outdoor time	Not Tested	
Gemini Copilot	Not Tested	
SACHET alerts	Not Tested	
NASA EONET	Not Tested	
Interactive map	Not Tested	
Citizen reports	Not Tested	
Recent history	Not Tested	
Multi-model comparison	Not Tested	
Marine mode	Not Tested	
Flood guidance	Not Tested	
Experimental nowcast	Not Tested	
Weather Route Intelligence	Not Tested	
Campus Mode	Not Tested	
Event Planner	Not Tested	
AI Trip Planner	Not Tested	
Scenario Simulator	Not Tested	
Heat Stress Intelligence	Not Tested	
Indian languages	Not Tested	
Voice	Not Tested	
Mobile view	Not Tested	
Backend offline handling	Not Tested	
API error handling	Not Tested	

10. Suggested Remote Working Method

1. Morning: agree on the day's targets in the group chat.
2. Prototype team builds and shares one clearly named latest build.
3. Research members send short findings and evidence to the PPT/documentation member.

4. Testing member tests the newest build and updates the bug list.
5. PPT member updates screenshots, diagrams and presentation content continuously.
6. Night: hold a 15-20 minute meeting to review completed work, blockers and next-day targets.

11. Team Rules to Avoid Integration Problems

- Only the prototype team should make regular edits to the core application files.
- Keep one master WeatherGPT build; do not circulate many final/final2/final-new ZIP files.
- Any backend response-format change must be communicated before frontend integration.
- Every member must document what they changed and what still remains incomplete.
- Features should be integrated and tested gradually, not combined for the first time one day before SIH.
- No member should claim an official warning, prediction accuracy or AI capability that has not been validated.

12. Final Member Allocation - Fill Here

Role	Member Name	Contact / Notes
Prototype - Frontend		
Prototype - Backend / AI		
Testing & Integration QA		
PPT & Documentation		
Research		
Product Research & SIH Strategy		

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